User Guide - indirect method.

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Suppose we have given a symmetric shift register with parameters k, p and n. The goal is to determine minimal periods in the general case. This will be done by a method based on the inductive method in menu option 1. In order to use the inductive method some assumptions must be fulfilled. The first step is therefore to reformulate the problem such that these conditions are fulfilled. Then the minimal period can be determined by the inductive method (menu option 1). This method is very fast and can compute very large periods. The note "The indirect method" gives a brief summary of the method being used.

In the descriptions we have slightly modified the notation regarding variable names in the computer program. For example, we write w_{min} and k_{new} instead of $minw$ and $knew$. The results and methods in this note are based on [1] "Representation of symmetric shift registers" by Jan Søreng, published and available at arXiv [2505.23974].

Table of contents.

Section 1. Notation.
Section 2. Vector representations.
Section 3. Input.
Section 4. Assumptions.
Section 5. Short description of the method.
Section 6. Output.
Section 7. Limitations.

1. Notation.

If $j \geq 1$, then 0_j denotes j consecutive 0's, and 1_j denotes j consecutive 1's. Moreover, if $j = 0$, then 0_j and 1_j denote the empty string.

We call $a_1 \cdots a_n$ a string if $a_i \in \{0, 1\}$ for $1 \leq i \leq n$. In the same way we also call $a_1 a_2 \cdots$ a string if $a_i \in \{0, 1\}$ for $i \geq 1$.

If $A = a_1 \cdots a_n$ is a string, then

$$w(A) = \text{the number of bits equal to 1 in } A = a_1 + \cdots + a_n.$$

2. Vector representations.

Suppose $Q = (q_1, \cdots, q_s)$ where s is even, $q_1 \geq 0$, $q_i > 0$ for $1 \leq i < s$, and $q_s \geq 0$. Then Q is the even vector representation of the string

$$A = 1_{q_1} 0_{q_2} 1_{q_3} 0_{q_4} \cdots 1_{q_{s-1}} 0_{q_s}.$$

Example 2.1. If $A = 110011100$, then we decompose $A = 1_2 0_2 1_3 0_2$ and get that $Q = (2, 2, 3, 2)$ is the even vector representation of A.

Example 2.2. If $A = 111100111$, then we decompose $A = 1_4 0_2 1_3 0_0$ and get that $Q = (4, 2, 3, 0)$ is the even vector representation of A.

Example 2.3. If $Q = (3, 4, 2, 0)$, then Q represents

$$A = 1_3 0_4 1_2 0_0 = 111000011.$$

Example 2.4. If $Q = (2, 3, 1, 3)$, then Q represents

$$A = 1_2 0_3 1_1 0_3 = 110001000.$$

Example 2.5. If $Q = (0, 3, 1, 0)$, then Q represents $A = 1_0 0_3 1_1 0_0 = 0001$.

Example 2.6. If $Q = (0, 3, 1, 5)$, then Q represents

$$A = 1_0 0_3 1_1 0_5 = 000100000.$$

3. Input.

A vector $Q = (q_1, \cdots, q_s)$ satisfying

$$s \text{ is even, } q_1 \geq 0, \ q_i > 0 \text{ for } 2 \leq i < s, \text{ and } q_s \geq 0,$$

must be entered into the first input field. The input vector should be comma separated and must be given without parenthesis. $n = q_1 + q_2 + \cdots + q_s$ is calculated. In the next input field the value of p should be entered. The p-value must satisfy $0 \leq p < n$.

In the third input field the value of k is entered. The k-value must satisfy

$$k_{min} \leq k \leq k_{max} \tag{3.1}$$

where k_{min} and k_{max} are calculated by the program.

In fact,

$$k_{min} = max\{0, w - (p+1\},$$
$$k_{max} = min\{w, n - (p+1)\},$$

where $w = q_1 + q_3 + \cdots + q_{s-1}$.

4. Assumptions.

Let $A = a_1 \cdots a_n$ be the string corresponding to Q. The condition (3.1) ensures that

$0 \leq k \leq w(A) \leq k + p + 1 \leq n$ where $w(A)$ is the weight of A.

We let $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ be the string generated from A by the symmetric shift register with parameters k, p and n. For $i \geq 0$, then

$a_{n+i+1} = a'_{i+1}$ if $k \leq a_{i+2} + \cdots + a_{i+n} \leq k + p$. Otherwise, $a_{n+i+1} = a_{i+1}$.

The minimal period of A^∞ is called the minimal period of A with respect to the symmetric shift register with parameters k, p and n. Let

$$w_i = a_{i+1} + \cdots + a_{i+n} - k \text{ for } i \geq 0. \tag{4.1}$$

Then $a_{i+1} + \cdots + a_{i+n} = w_i + k$ for $i \geq 0$. We call $w_0, w_1, \cdots$ the weight parameters of A with respect to k, p and n.

Let $A_i = a_{i+1} \cdots a_{i+n}$ for $i \geq 0$. In particular, $A_0 = A$. We note that

$$w_i = a_{i+1} + \cdots + a_{i+n} - k = w(A_i) - k \text{ for } i \geq 0. \tag{4.2}$$

Since $k \leq w(A) \leq k + p + 1$, then by Part 6 in [1] we get that

$$k \leq w(A_i) \leq k + p + 1 \text{ and } 0 \leq w_i \leq p + 1 \text{ for } i \geq 0. \tag{4.3}$$

5. Short description of the method.

The program calculates

$$w_{min} = min\{w_i : 0 \leq i \leq 4n\} \text{ and } w_{max} = min\{w_i : 0 \leq i \leq 4n\}.$$

Then it can be proved that

$$min\{w_i : i \geq 0\} = w_{min} \text{ and } max\{w_i : i \geq 0\} = w_{max}.$$

Let $p_{new} = w_{max} - w_{min} - 1$ and $k_{new} = k + w_{min}$. Then it can be proved that $a_1 \cdots a_n a_{n+1} \cdots$ is also generated from A by the symmetric shift register with parameters k_{new}, p_{new} and n.

Let $w_i = w_i - w_{min}$ for $0 \leq i \leq 4n$. Then it can be proved that $w_0, w_1, \cdots, w_{4n}$ are weight parameters of A with respect to k_{new}, p_{new} and n.

Let $s \geq 0$ be minimal such that $w_s = p_{new} + 1$. Let $t > s$ be minimal such that $w_t = 0$. Let r be maximal such that $s \leq r < t$ and $w_r = p_{new} + 1$.

Let $A_{new} = a_{r+1} \cdots a_{r+n}$. Let Q_{new} be the even vector representation of A_{new}. Next, Q_{new} is used as the input vector to the inductive method (menu option 1). It can be proved that p_{new} is one of the p - values that can be selected. We use p_{new} as p - value and we get by the inductive method the minimal period of A_{new} with respect to the symmetric shift register with parameters k_{new}, p_{new} and n. By [1] this will also be the minimal period of A with respect to the symmetric shift register with parameters k, p and n.

6. Output.

Auxiliary parameters: w_{min}, w_{max}, r, s and t.

Main results: $p_{new}, k_{new}, Q_{new}$ and the minimal period.

We refer to Section 5 for an interpretation of these results. In most of the cases $p_{new} = p$ and $k_{new} = k$. The first two examples are included to illustrate that p_{new} and k_{new} can be different from p and k. The last example demonstrates the strength of the indirect method. In this case the minimal period is very large and it will be very time consuming if we use the direct method.

Example 6.1.

Input: $Q = 2, 1, 2, 1, 2, 1, 2, 2 \qquad p = 10 \qquad k = 1.$

Auxiliary results: $w_{min} = 2$, $w_{max} = 9$, $r = 24$, $s = 24$ and $t = 37$.

Main results: $p_{new} = 6$, $k_{new} = 3$, $Q_{new} = (4, 1, 2, 1, 2, 1, 2, 0)$
and minimal period = 26.

Example 6.2.

Input: $Q = 2, 3, 8, 5, 4, 0 \qquad p = 10 \qquad k = 10.$

Auxiliary results: $w_{min} = 0$, $w_{max} = 5$, $r = 5$, $s = 5$ and $t = 10$.

Main results: $p_{new} = 4$, $k_{new} = 10$, $Q_{new} = (8, 5, 4, 2, 3, 0)$
and minimal period = 112.

Example 6.3.

Input: $Q = 3, 100, 4, 200, 300, 600 \qquad p = 45 \qquad k = 270.$

Auxiliary results: $w_{min} = 0$, $w_{max} = 46$, $r = 307$, $s = 15$ and $t = 353$.

Main results: $p_{new} = 45$, $k_{new} = 270$, $Q_{new} = (300, 603, 12, 92, 4, 196)$
and minimal period = 1 731 106 405.

7. Limitations.

Since the indirect method is based on the inductive method, the limitations of the inductive method will also apply to the indirect method. These re-

strictions are due to the fact that the values of the integer variables must remain within certain limits.

Due to technical constraints the inductive method may produce incorrect results if the periods exceed Long.MaxValue (92233720366854775807). Therefore the method works if p and the coordinates of Q are not too large. It is difficult to predict the size of the calculated periods. Hence, it is not easy to say anything precisely about how large Q and p can be. But examples with periods equal to about 100 000 000 have been tested to be correct by using the definition of the shift register step by step.

However, we think that the program will give correct answers for much larger periods. If we suspect that the answers are wrong, we can use the inductive method in menu option 1 with Q_{new} as input and with p_{new} as p-value. Then some of the intermediate calculations are displayed. Then we can assess whether these calculations seem reasonable.